Status	Finished
Started	Saturday, 10 October 2026, 6:50 PM
Completed	Saturday, 10 October 2026, 6:57 PM
Duration	6 mins 50 secs
Grade	**12.00** out of 15.00 (**80**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O👎
- b. O(log n) ✓
- c. O(1)
- d. O(n log n)

Question **2**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The higher middle element is chosen as the middle element
- b. The search stops
- c. The lower middle element is chosen as the middle element ✗
- d. The middle element is chosen randomly

Question **3**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is somewhere in the middle of the array ✓
- b. Item is the last element in the array or item is not there at all
- c. When the item is the last element in the array
- d. When the item is not the array at all

Question **4**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O👎 ✓
- b. O(n log n)
- c. O(1)
- d. O(log n)

Question **5**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Binary search ✓
- b. Hash search
- c. Linear search
- d. Sequential search

Question **6**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 10 ✓
- b. 100
- c. 500
- d. 1000

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search requires the list to be sorted.
- b. Linear search divides the list into halves.
- c. Linear search can be applied to both sorted and unsorted lists. ✓
- d. Linear search is more efficient than binary search on large lists.

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Linear search ✓
- b. Hashing
- c. Dynamic search
- d. Binary search

Question **9**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Binary search
- b. Hash search
- c. Linear search ✓
- d. Both (1) & (3)

Question **10**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

- a. Compare the target element with the middle element in the list
- b. Sort the list ✗
- c. Compare the target element with the first element in the list
- d. Divide the list into two equal parts

Question **11**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It works on unsorted lists
b. It can be applied only if the list is sorted ✓
c. It compares elements sequentially
d. It always starts from the beginning of the list

Question **12**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Searching ✓
b. Mining
c. Finding
d. Discovering

Question **13**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search continues until the end of the list
b. The search stops immediately ✓
c. The search starts over from the beginning
d. The search backtracks to find duplicate elements

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Linear search ✓
b. Quick search
c. Merge search
d. Bubble search

Question **15**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99 ✕

b. 89 and 94

c. 90 and 94

d. 90 and 99